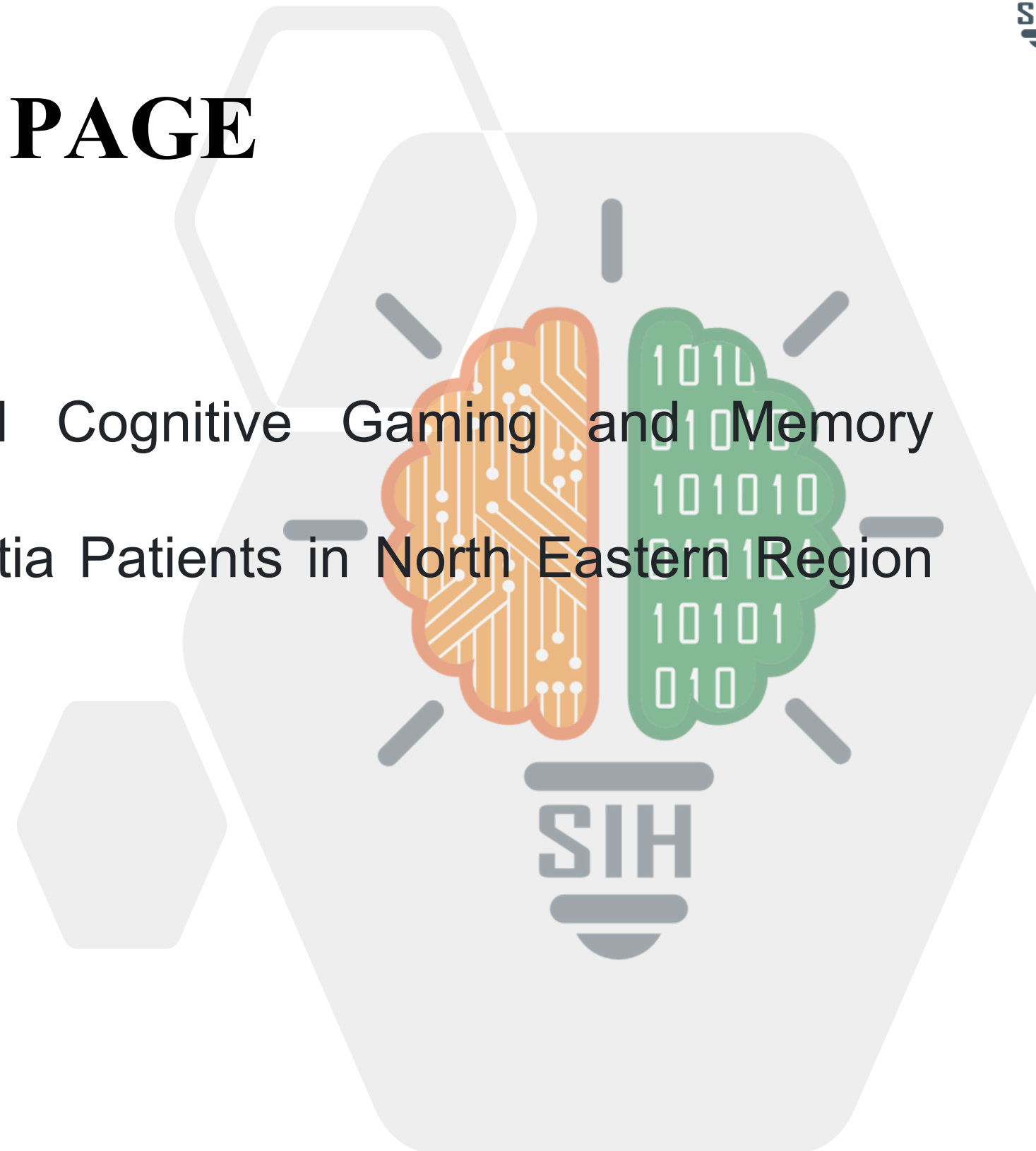


## TITLE PAGE

- **Problem Statement ID-** 26003
- **Problem Statement Title-** AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-**
- **Team Name-** OneCommitAway

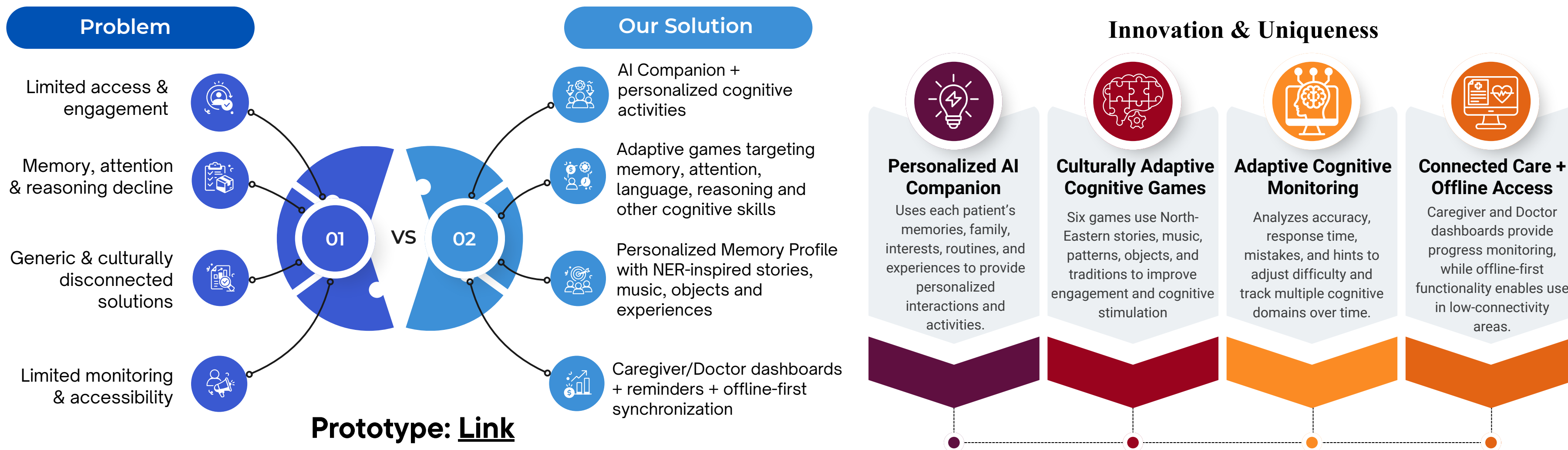




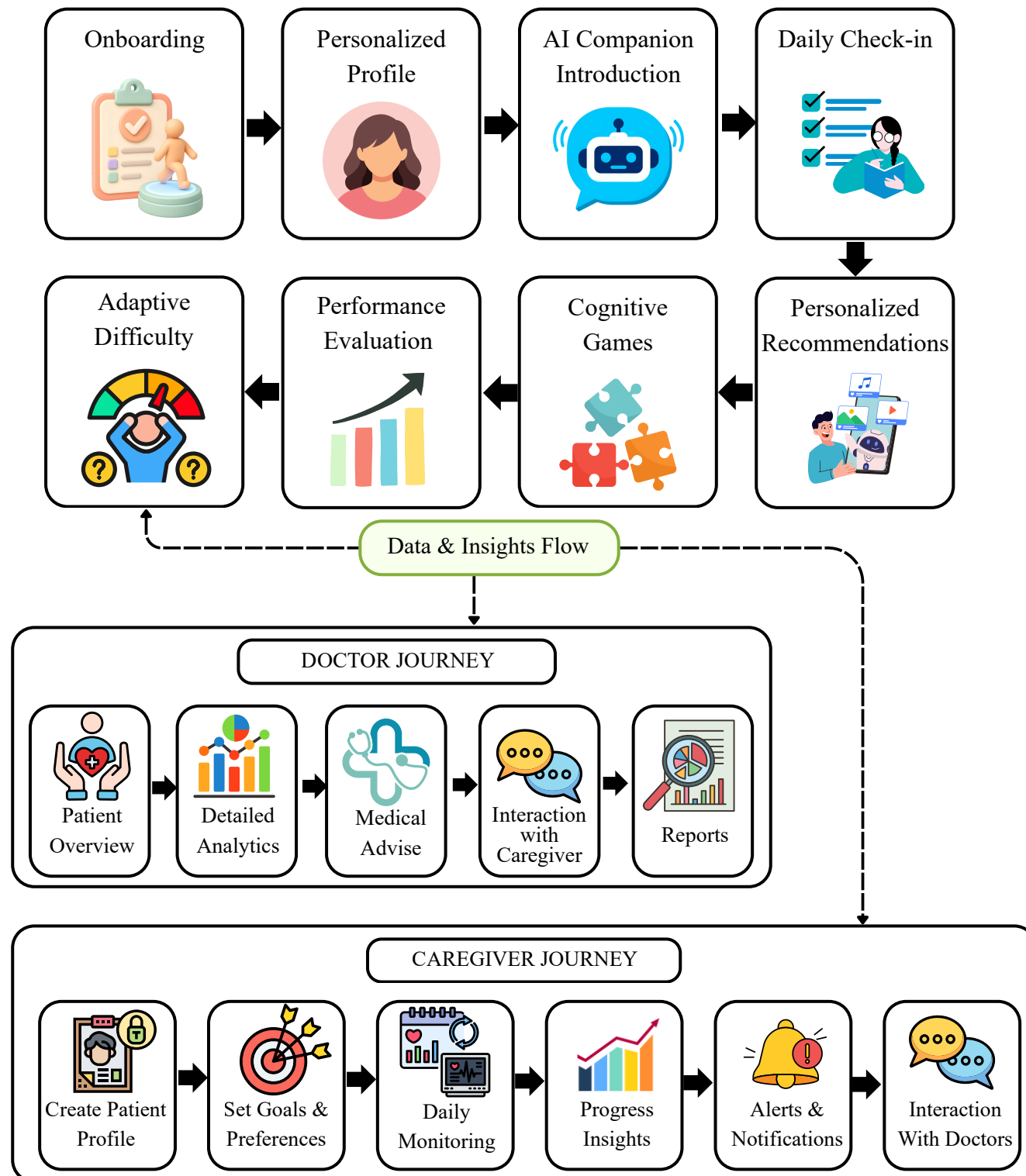
# IDEA TITLE

## Detailed Explanation of the Proposed Solution

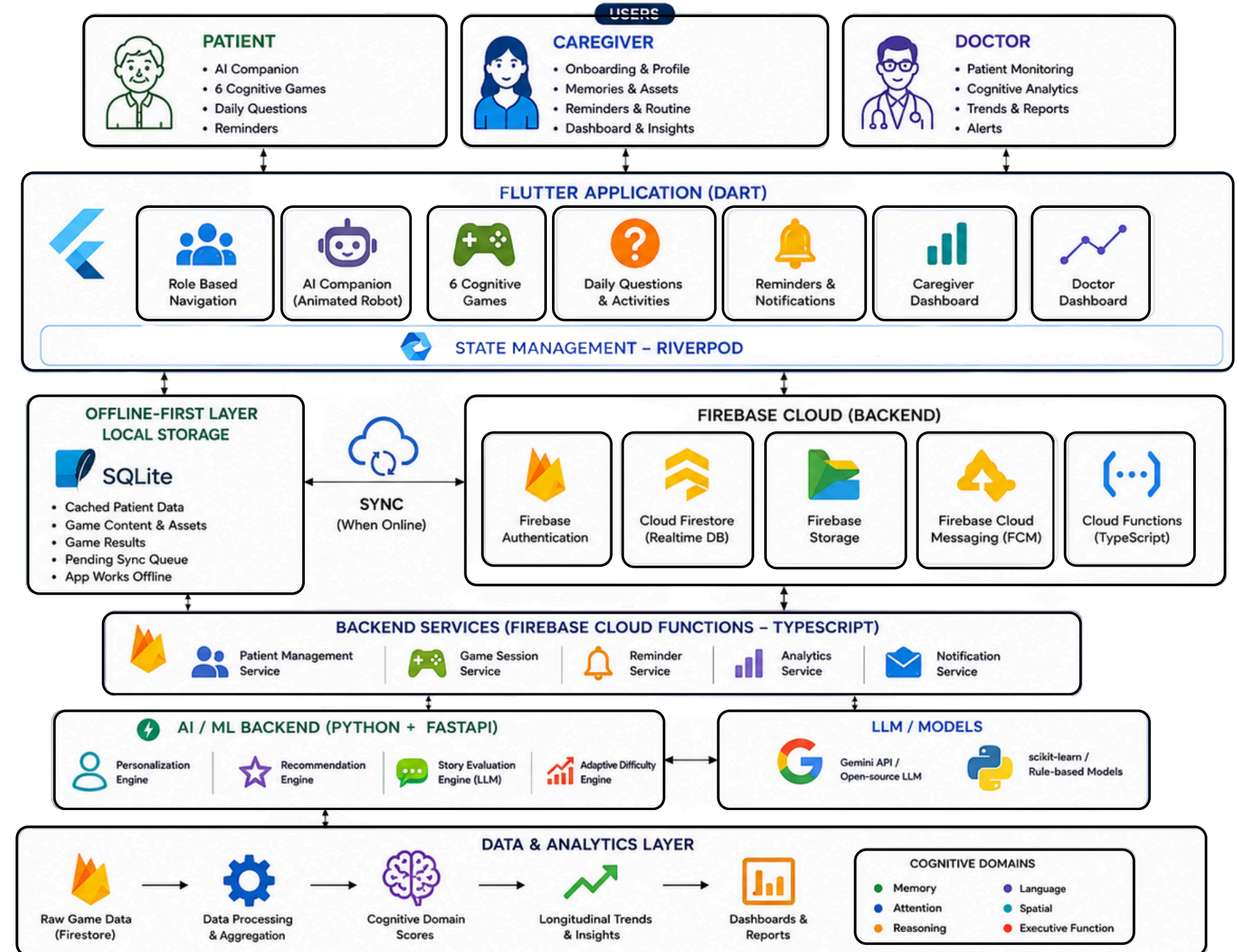
We propose an **AI-powered cognitive assistance platform for elderly dementia** patients with three roles: **Patient, Caregiver, and Doctor**. Caregivers create personalized profiles using the patient's memories, interests, routines, photos, and familiar experiences. An **AI Companion** then provides personalized conversations, reminders, and activities. The platform offers **six adaptive cognitive games** covering memory, attention, reasoning, language, spatial, auditory, and procedural skills, using **North-Eastern culture and familiar content** for engagement. Performance data helps the system **adapt difficulty and recommend activities**. Caregivers track progress through dashboards, while doctors monitor **long-term cognitive trends and alerts**. The platform is **Multilingual, voice-enabled, and offline-first** for easy access in familiar languages, even in low-connectivity areas.



Flow Diagram

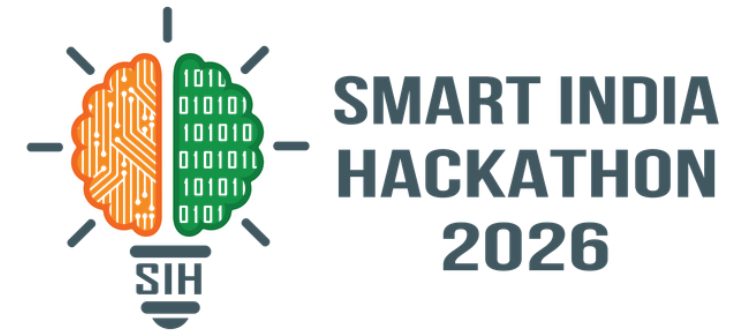


Architecture Diagram

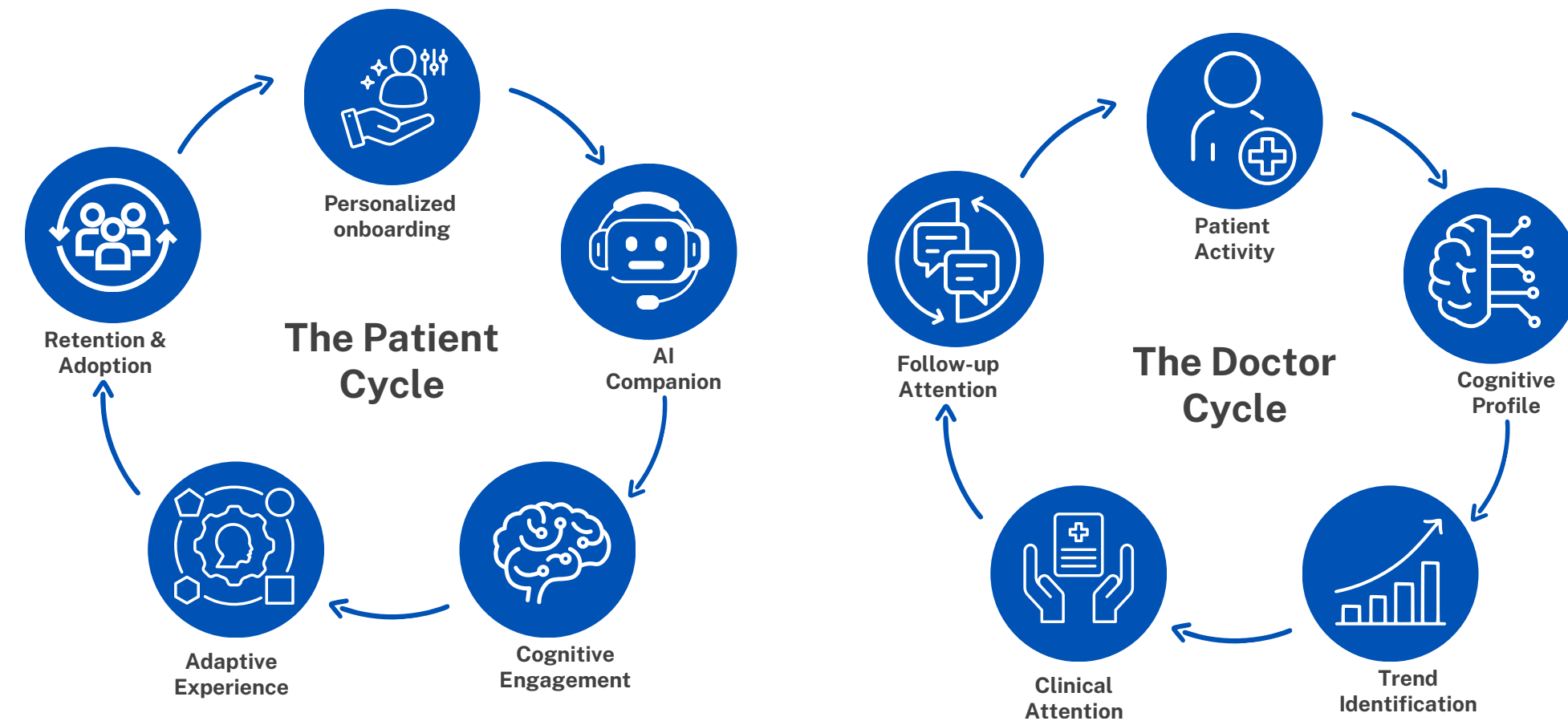




# FEASIBILITY AND VIABILITY



## Viability



Prototype: [Link](#)

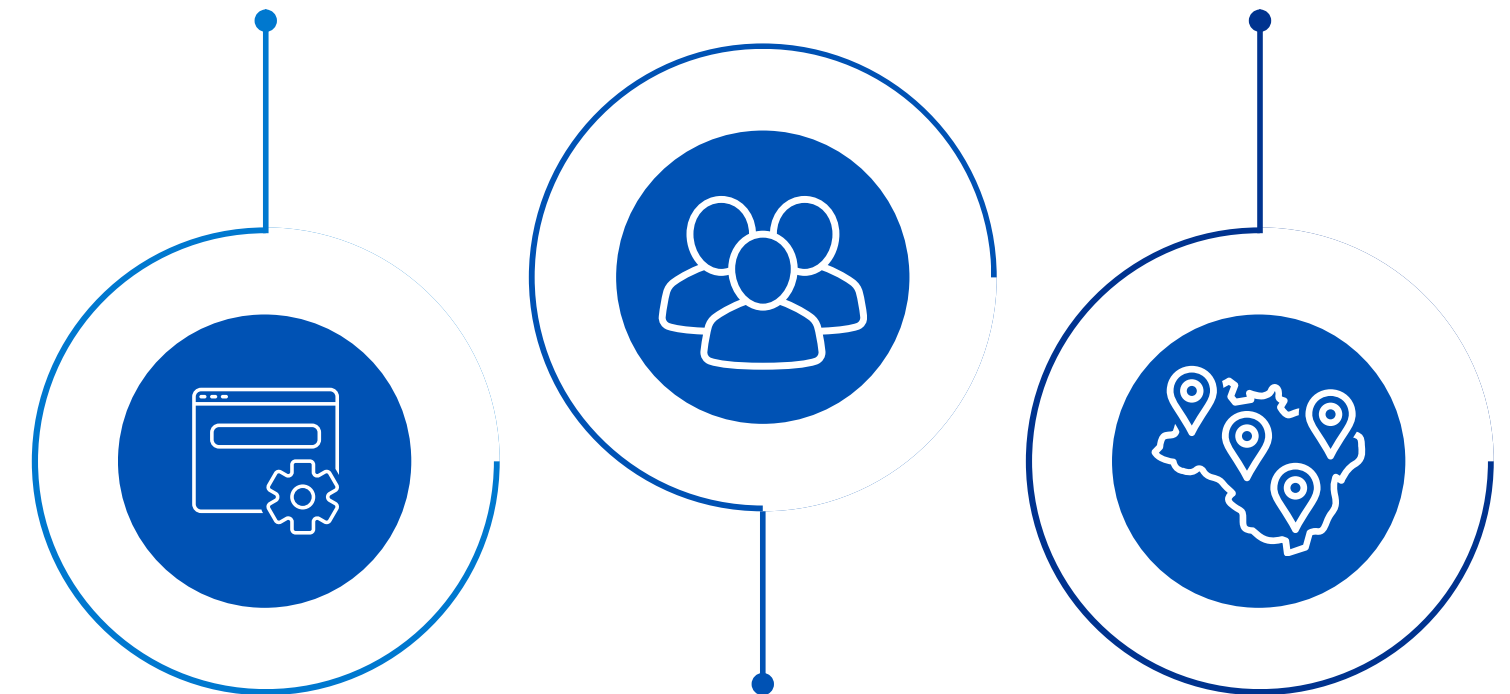
## Feasibility

### Technical Feasibility

- Developed using Flutter for cross-platform support
- AI models for cognitive assessment
- Cognitive games to engage and assess users

### Regional Feasibility

- Works offline with local data storage
- Background sync when connectivity is available
- Supports remote regions across NER



### User Feasibility

- Voice-first interface for ease of interaction
- Caregiver-assisted setup and monitoring
- Designed for elderly users with limited digital literacy



# IMPACT AND BENEFITS



SMART INDIA  
HACKATHON  
2026



## Impacts

### Improved Cognitive Engagement (Patient)

AI-personalized activities and adaptive difficulty levels encourage regular cognitive engagement.

### Reduced Caregiving Burden (Caregiver)

Activity tracking, reminders, and progress visibility helps monitor the patient's engagement.

### Better Clinical Context (Doctor)

Engagement data provide doctors with a clearer view of the patient's progress before consultation.

### Improved Access to Cognitive Support

An offline-first platform extends continuous cognitive engagement for elderly dementia patients, including those in remote NER regions.

01

02

03

04

01

02

03

04

## Benefits



### Personalized Experience (Patient)

Personalized care for every patient based on their memories, interests, routines, and abilities.

### Easy Monitoring & Support (Caregiver)

Caregivers can track activity completion, engagement trends and reminders through a simple dashboard.

### Time-Efficient Decision Support (Doctor)

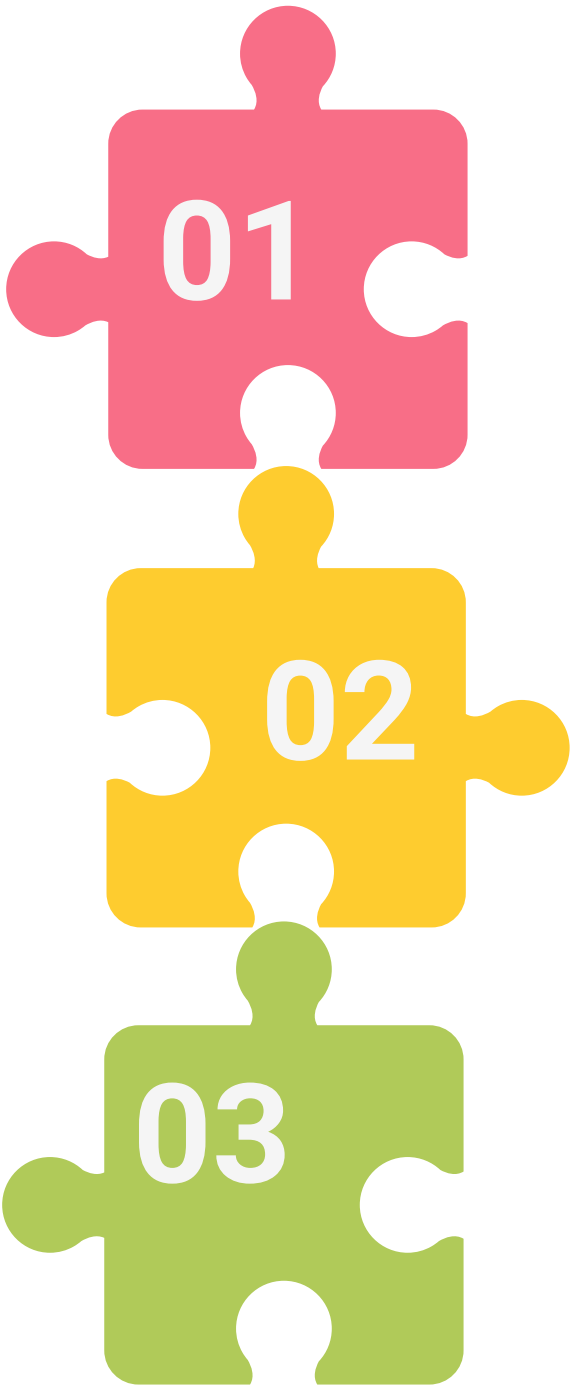
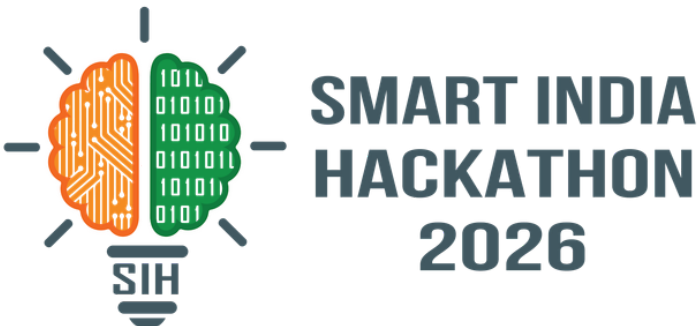
Longitudinal performance insights provide relevant patient context before consultations.

### Sustainable & Scalable Care Model

The integration of patients, caregivers, doctors and rural onboarding support creates a connected model that can expand access beyond individual users.



# RESEARCH AND REFERENCES



**1.Prevalence and Correlates of Dementia among the Community- Dwelling Elderly of Guwahati City, Assam | [Link](#)**

- To assess the prevalence of dementia among elderly residents of Guwahati and identify its associated demographic, socioeconomic and lifestyle factors.

**2.A Community-Based Study on Functional Disability and Its Associated Factors among Elderly Individuals in a Rural Setting in Northeastern India | [Link](#)**

- To assess the prevalence of functional disability among rural elderly and identify factors associated with difficulty in daily activities.

**3.Depression and Its Determinants among Elderly Living in Dibrugarh District, Assam: A Community-Based Cross-Sectional Study| [Link](#)**

- To determine the prevalence of depression among elderly in Dibrugarh and identify its associated social, physical and health-related factors.

**Literature Survey Document: [Link](#)**

**Prototype: [Link](#)**

Existing Solution

| FEATURES             | COGNIVERSE | SMART4MD | ARDSI GUWAHATI | MEMORYMITRA |
|----------------------|------------|----------|----------------|-------------|
| Cognitive Games      | ✓          | ✓        | ✓              | ✓           |
| Memory Assistance    | ✗          | ✓        | ✓              | ✓           |
| Adaptive Difficulty  | ✓          | ✗        | ✗              | ✓           |
| Caregiver Monitoring | ✓          | ✓        | ✗              | ✓           |
| Voice & Multilingual | ✓          | ✗        | ✗              | ✓           |
| NER Cultural Content | ✗          | ✗        | ✗              | ✓           |